The Cutting Edge and Future for Neural Networks and Deep Learning

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University of Colorado Boulder
Spring 2026



https://dannagurari.colorado.edu/course/neural-networks-and-deep-learning-spring-2026/

Review

- Last lecture:
  - Acceptability game: judging ethics of NN development and use decisions
- Assignments (Canvas):
  - Project presentations due in 1.5 weeks
  - Peer evaluation in 2 weeks
- Next four lectures: Join via Zoom and then Gather (NOT IN-PERSON)
- Questions?

Final Projects: Online “Conference”

- Gather platform

- Peer evaluation: each person must provide 3 types of feedback for assigned room
  - A rose: What do you think was the highlight or biggest success of this project?
  - A thorn: What was a challenge or weakness in their project that you think needs to be addressed?
  - A bud: What are the implications of their project for future research?

Today’s Topics

- The cutting edge of neural networks and deep learning
- How might future neural networks and deep learning look?
- How might future societies look?

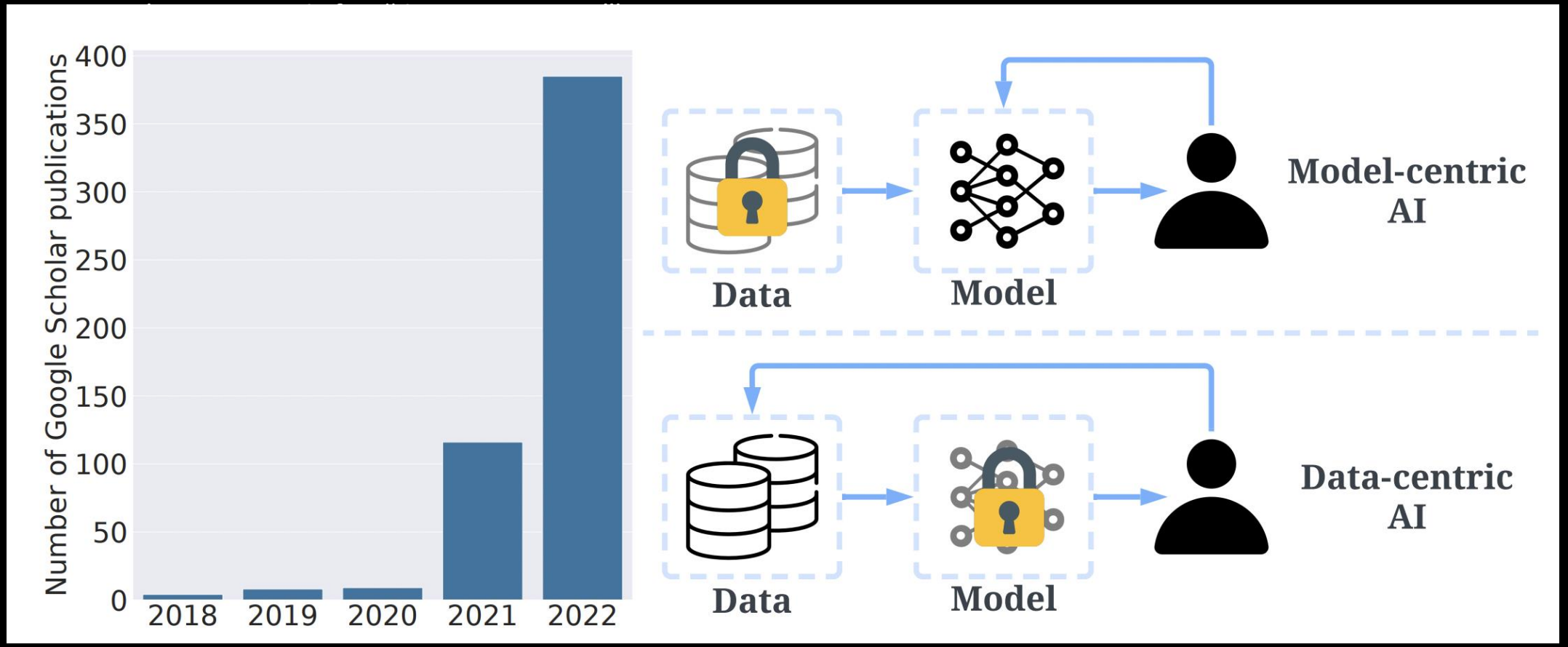
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Recall from Prior Lectures

- Multimodal generalist models: text, images, audio, videos, etc
- Efficiency tricks: bigger isn’t necessarily better
- Human alignment tricks

From Model-Centric to Data-Centric AI (1 of 5)



Greater focus on collecting data than designing models (e.g., GPT-1 to 4 series)

Zha et al. Data-centric AI: Perspectives and challenges. SDM 2023

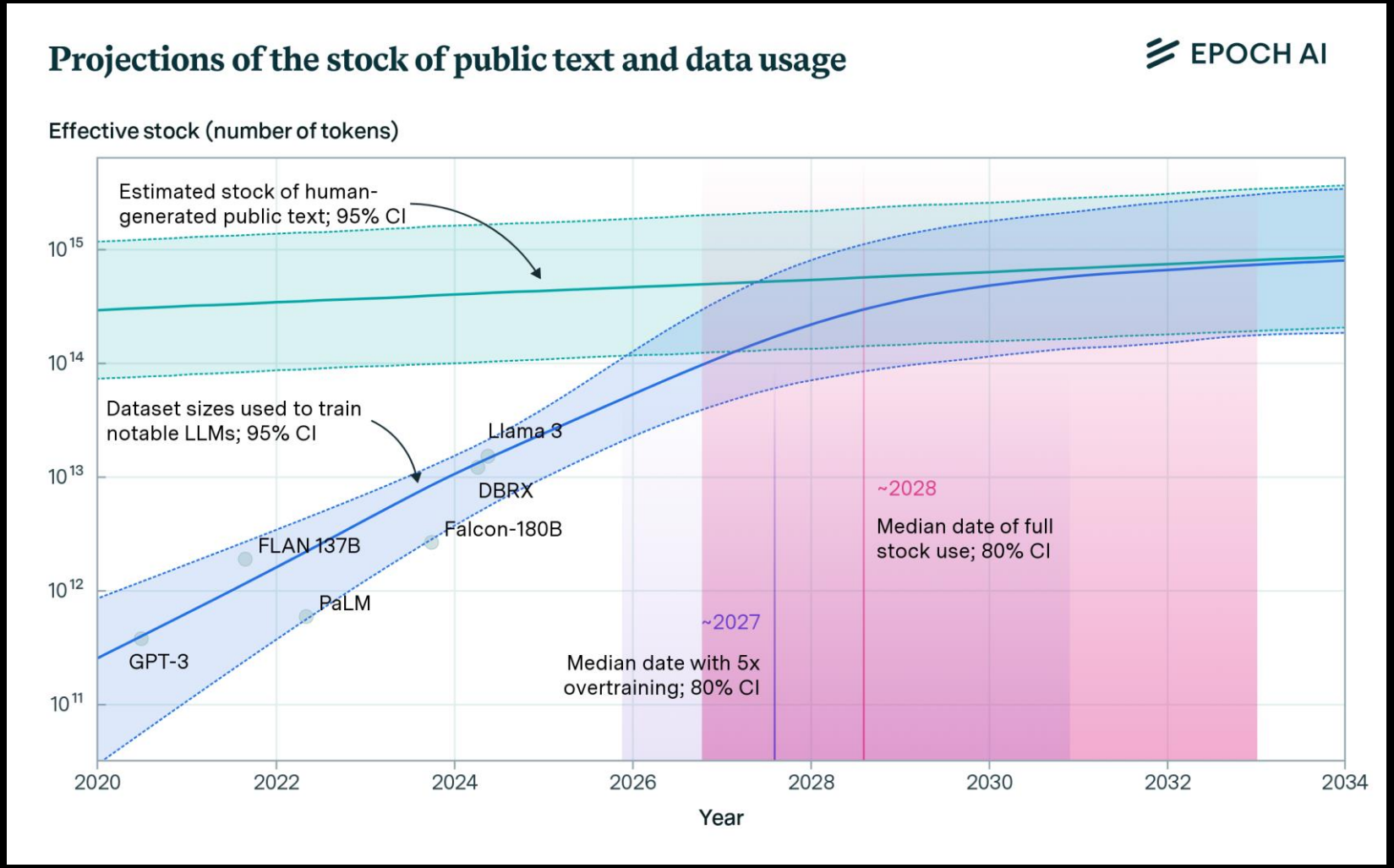
From Model-Centric to Data-Centric AI (1 of 5)

Problem: all public text data expected to be used some time between 2026 and 2032

Idea 1: synthetic training data

Idea 2: coreset selection (subset representative of dataset);

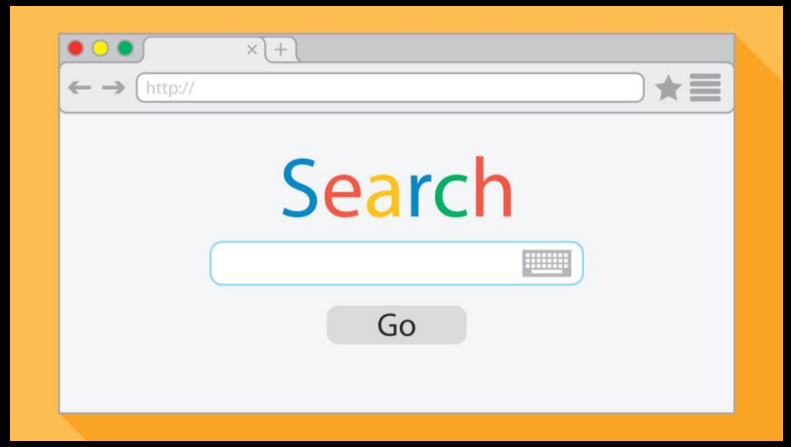
- Media-catching example in Feb 2025 is s1; it outperformed OpenAI's o1
- Qwen2.5 base model fine-tuned on 1,000 selected questions and corresponding Gemini-generated answers, which cost ~$20 to $50 in cloud compute credits.

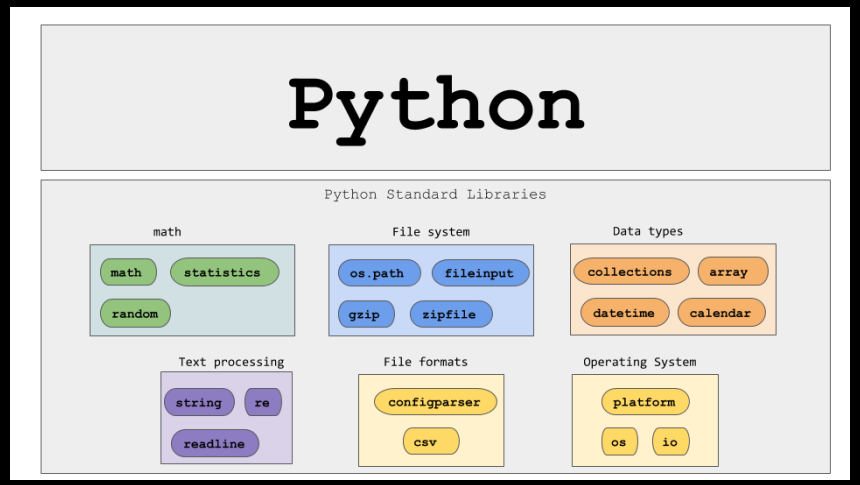


https://epoch.ai/blog/will-we-run-out-of-data-limits-of-llm-scaling-based-on-human-generated-data

Agentic AI (2 of 5)

Approach: moving beyond static prediction to autonomous decision-making systems that use external tools, analogous to humans' tool use; e.g., search, APIs



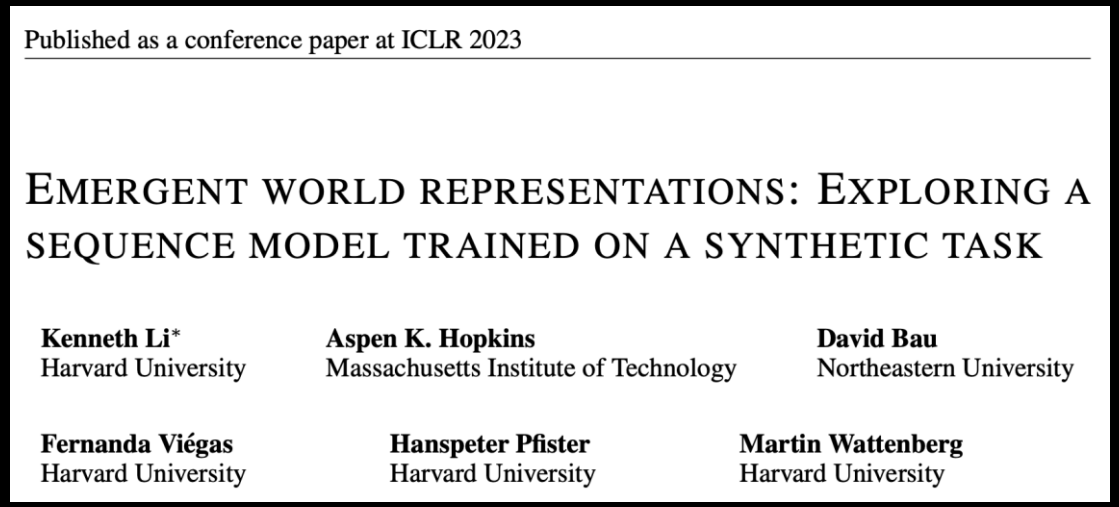


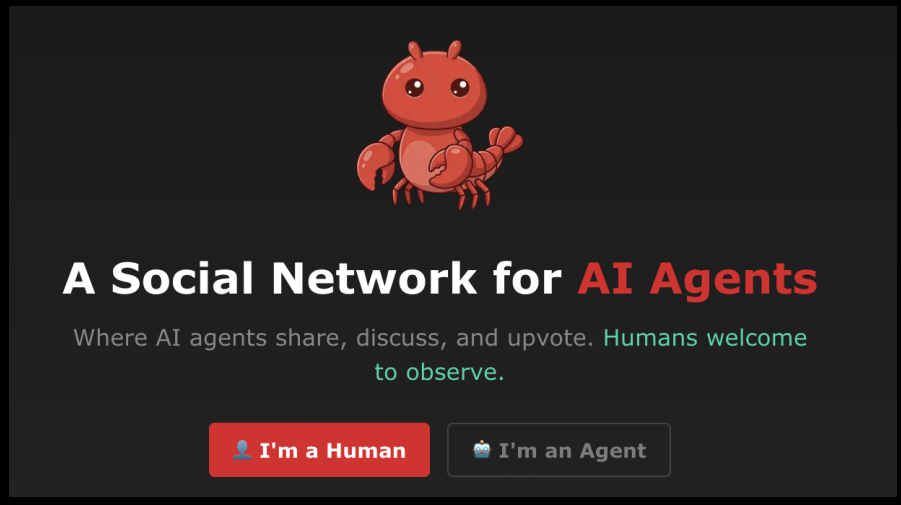
- Many concerns:
  - When should agents be vs not be allowed (e.g., government decision making, vacation bookings, meeting scheduling, shopping)?

Model Interpretability (3 of 5)









https://www.moltbook.com/

Neural networks largely remain scientific artifacts we grow and then try to understand, similar to humans

World Models (4 of 5)

- Idea: models with internal representations capturing rules of physical worlds
  - 2D video generators; e.g., Google's Project Genie (doesn't guarantee a consistent world since content hallucinated on the fly)
  - 3D world simulators; e.g., World Labs founded by Fei Fei Li
  - New architectures; e.g., Advanced Machine Intelligence founded by Yann LeCun inspired by his Joint Embedding Predictive Architectures (JEPAs)

Evaluation: Bridging Gap Between Benchmarks and Real-World Utility (5 of 5)

- Long tail of errors, rather than typical/average scenario
- Decisions that are context-specific, rather than universal

Today’s Topics

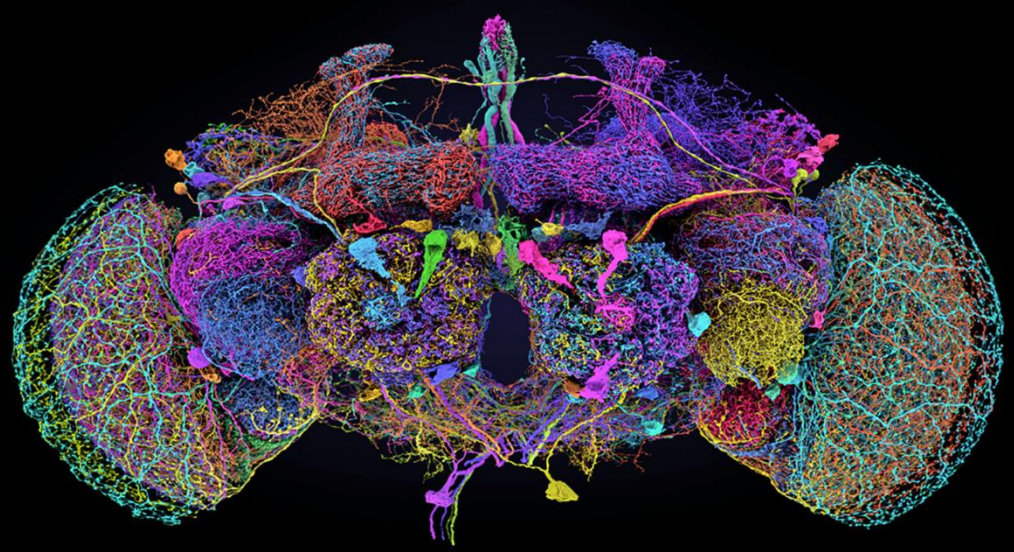
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Biology-Inspired Designs? We Are Actively Learning About How Brains Work

Complete map of fruit fly brain circuitry unveiled

AI tools and hundreds of human proofreaders helped tease apart 140,000 neurons in a brain the size of a grain of sand

2 OCT 2024 · 11:05 AM ET · BY RODRIGO PÉREZ ORTEGA



The 50 largest neurons in the fly's brain, shown here, are part of the largest, most complete neuronal map ever produced by scientists. TYLER SLOAN AND AMY STERLING FOR FLYWIRE/PRINCETON UNIVERSITY; DORKENWALD *ET AL.*, *NATURE*, 2024

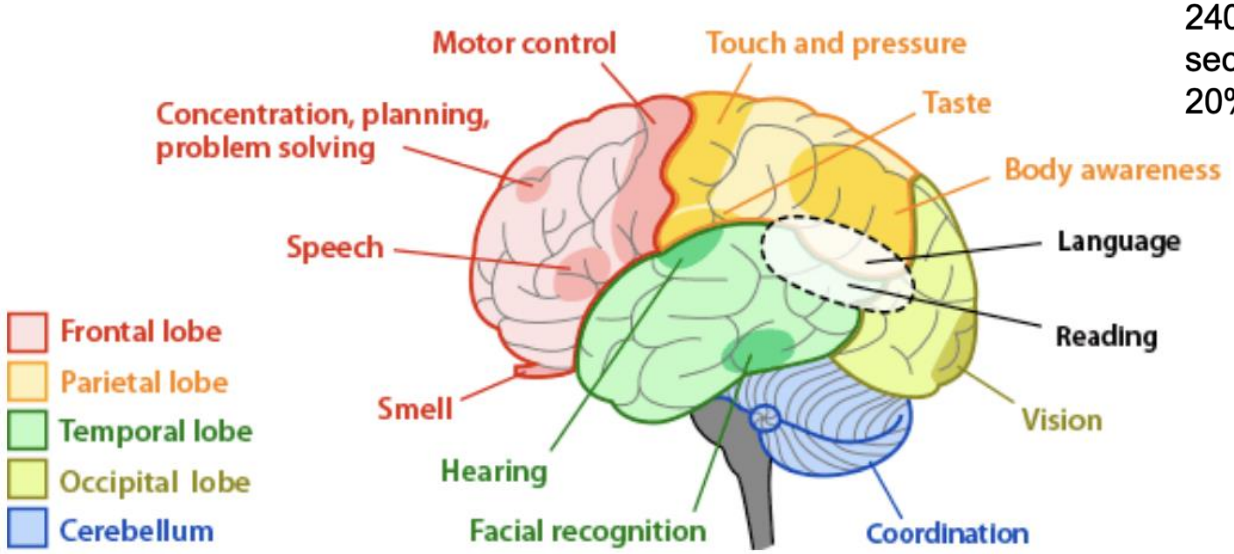
Browse yourself: https://codex.flywire.ai/app/view_3d

Prior work mapped nervous systems of a worm (~300 neurons) , fly larva (3,016), and most recently fruit fly (~140k, with 55M connections for an adult… that can fly, eat, fight, flee, and more!)

Future work will likely move to mice and eventually to humans

Biology-Inspired Designs? We Are Actively Learning About How Brains Work

- The computer as a brain that comprises **specialized accelerators**
- **Low power** – the brain consumes only about 20W
- **Fault tolerant** – the brain loses neurons all the time



2400 kcal/24 hr = 100 kcal/hr = 27.8 cal/sec = 116.38 J/s = 116 W
20% x 116 W = 23.3 W

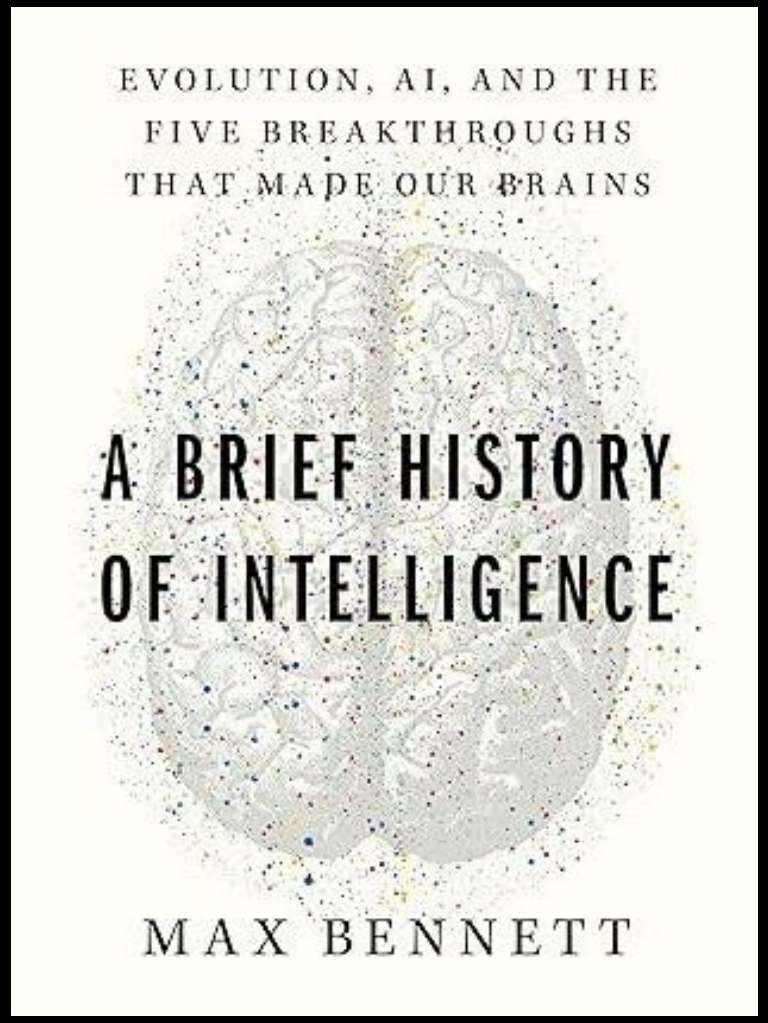
Yang, Eric. Think Dinner. *Mac Evolution, 1998*

https://askabiologist.asu.edu/sites/default/files/resources/articles/nervous_journey/brain-regions-areas.gif

Animals vs Neural Networks

- What does an animal need in order to grow and survive?
  - Habitat with sufficient air, water, and food (e.g., rotting food for flies)
- What does a modern NN system need in order to grow and survive?
  - More power than needed to support small cities
  - Massive amounts of data
- What is needed to bridge the gap between animals and NNs? e.g.,
  - Run at exaflop speed (a billion billion calculations per second) with 20W (current models need ~8 MW, or 8 million watts)
  - Simultaneously support many senses/modalities
  - Learn without backpropagation
  - Continuously learn, adapting and overcoming from injuries

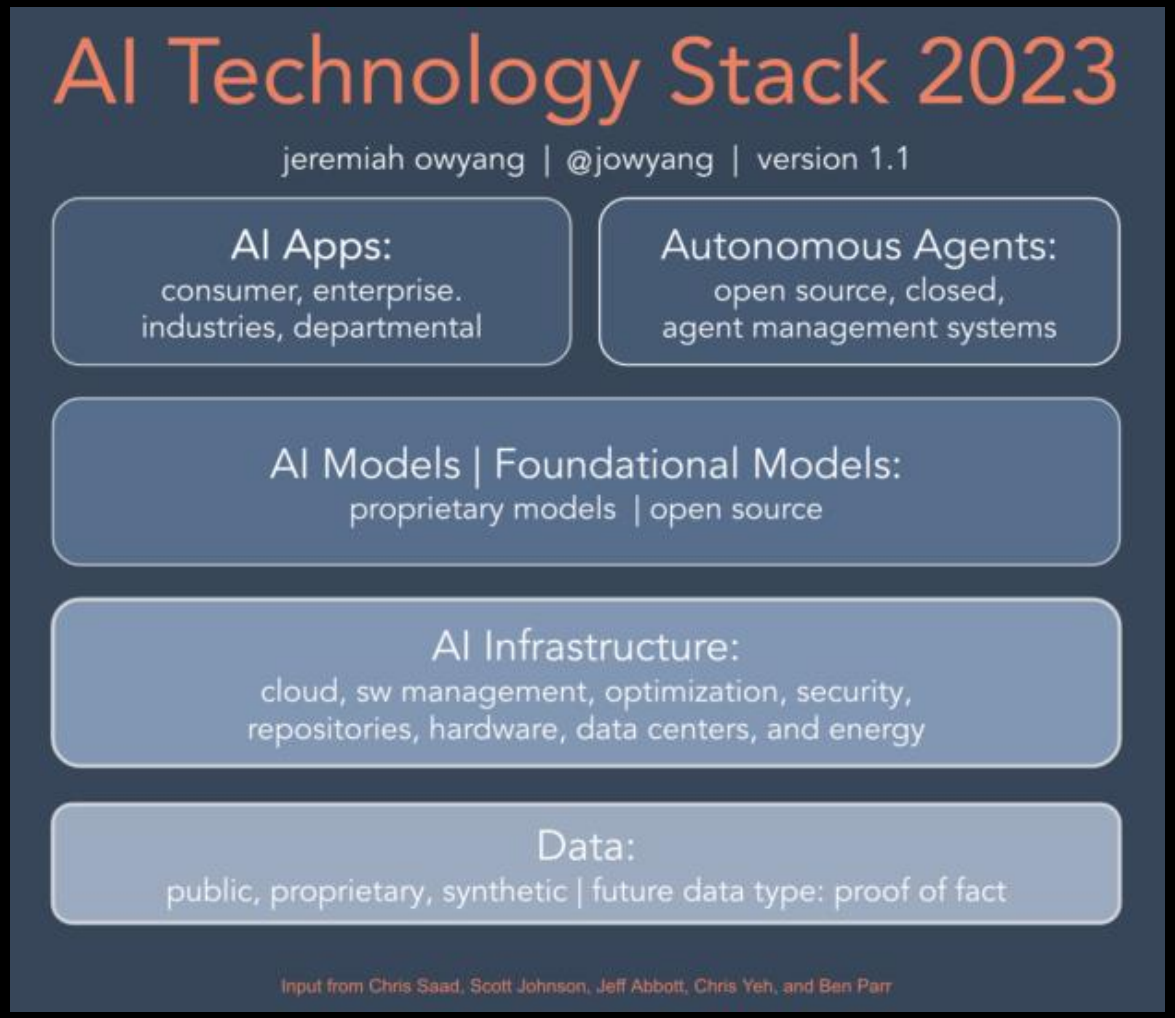
Biology-Inspired Designs: There is Still Much to Learn About How Brains Work



"The human brain contains eighty-six billion neurons and over a hundred trillion connections. Each of those connections is so minuscule—less than thirty nanometers wide—that they can barely be seen under even the most powerful microscopes. These connections are bunched together in a tangled mess—within a single cubic millimeter (the width of a single letter on a penny), there are over one billion connections.

But the sheer number of connections is only one aspect of what makes the brain complex; even if we mapped the wiring of each neuron we would still be far from understanding how the brain works. Unlike the electrical connections in your computer, where wires all communicate using the same signal—electrons—across each of these neural connections, hundreds of different chemicals are passed, each with completely different effects. The simple fact that two neurons connect to each other tells us little about what they are communicating. And worst of all, these connections themselves are in a constant state of change, with some neurons branching out and forming new connections, while others are retracting and removing old ones. Altogether, this makes reverse engineering how the brain works an ungodly task."

Holistic Perspective in Development: HW to SW



We have focused in lectures on ways models will improve via SW, improving their learning and efficiency

We will also see improvements to HW, potentially with:

- Quantum computing for faster computation
- Memristors that preserve a non-binary state (e.g., analog) after current/voltage passes through it, and so simultaneously support processing and storage (current devices use different modules)

It is faster to make changes in SW than HW

https://web-strategist.com/blog/2023/06/12/ai-technology-stack-2023-v1-1/

How Might Interaction Look with Models?

- Prompts aren’t always natural interactions
- What are more natural/desired ways to interface with machines?
  - Gestures
  - Physically (e.g., tactiley pointing to a screen or object)
  - Brain-computer interfaces
  - Combinations of multiple interfaces
  - Constraint-based graphical user interfaces (e.g., menus)

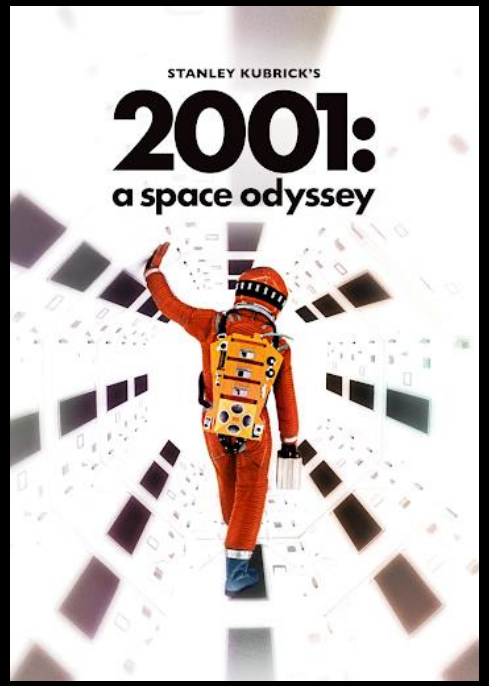
https://cacm.acm.org/opinion/prompting-considered-harmful/

Today’s Topics

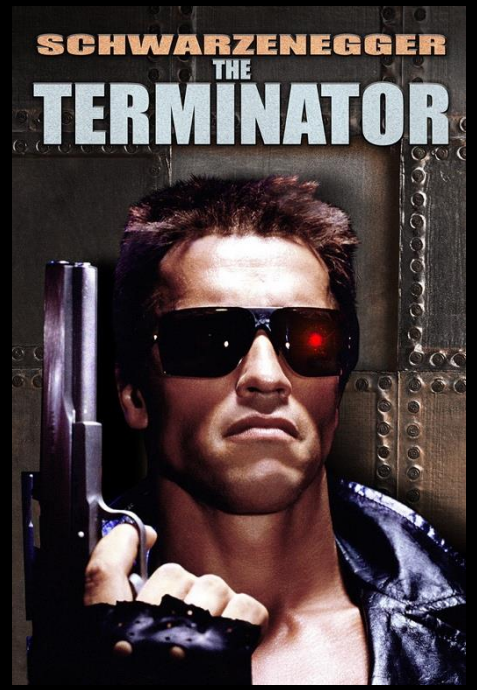
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What Entertainment Imagined

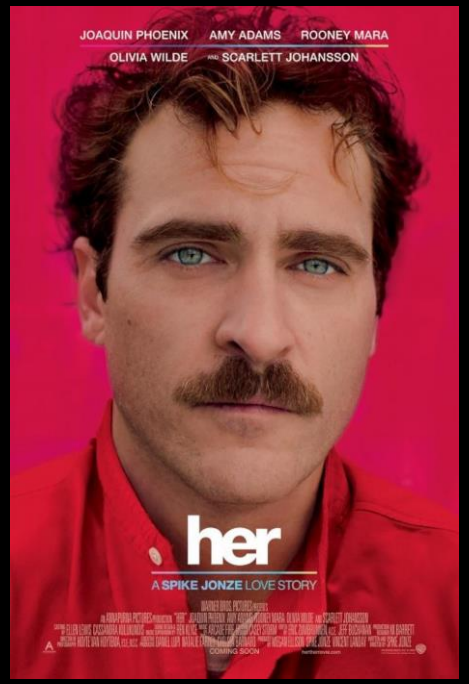
AI Goes Against Humans' Intents



AI Tries to Eliminate Humanity



AI is Intimate Partner (In Love & Friendship)



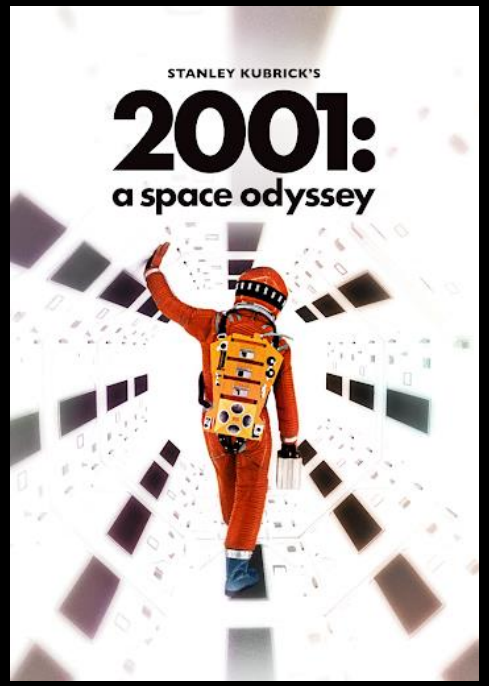
AI Enslaved, Even When Conscious & Feeling



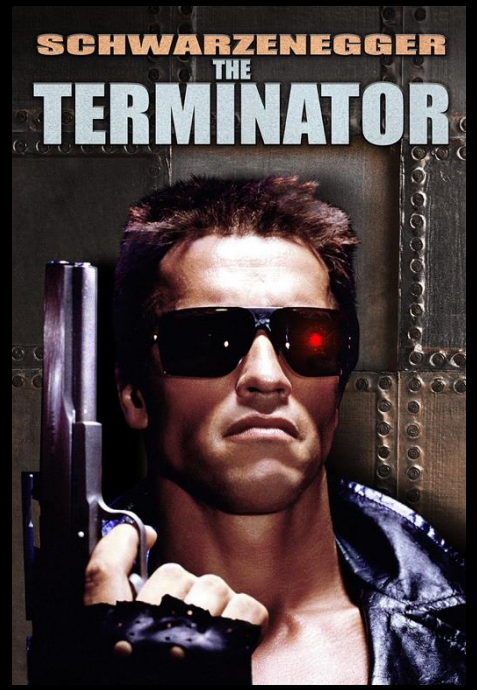
1968 1984 2013 2015

What Do You Think?

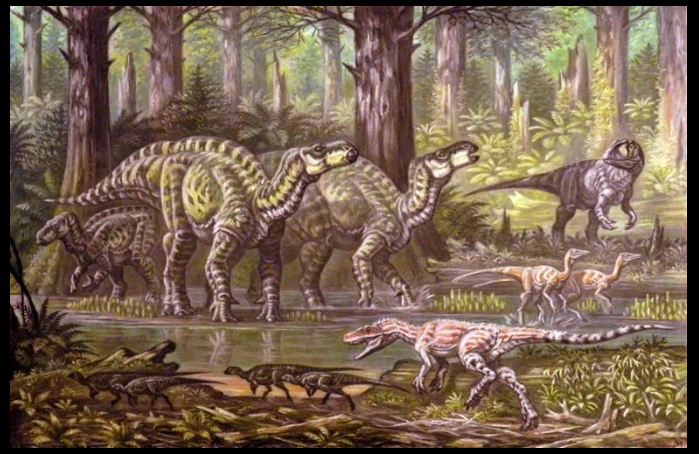
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AI Tries to Eliminate Humanity

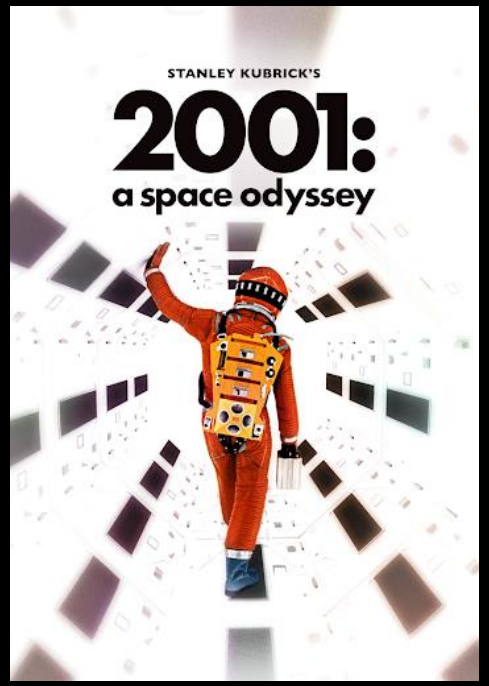


Will AI be an existential threat to humanity?

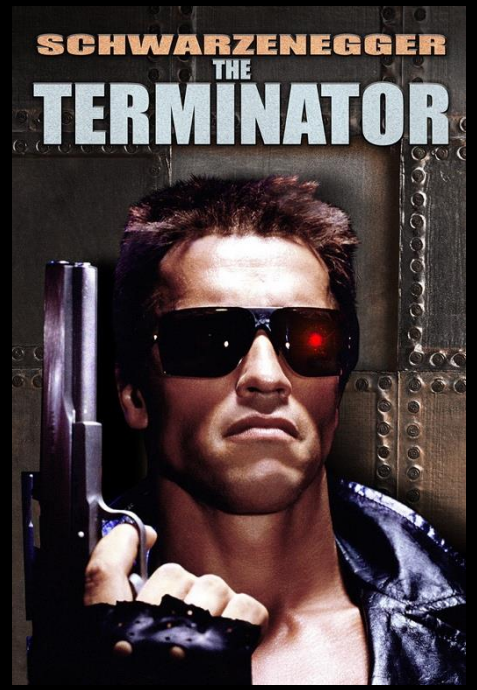


What Do AI Leaders Think?

AI Goes Against Humans' Intents



AI Tries to Eliminate Humanity



Will AI be an existential threat to humanity?



"10-20% chance they'll wipe us out"

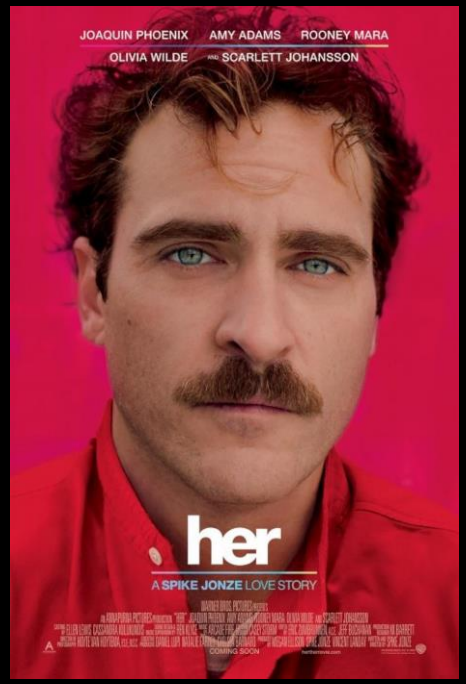
Geoffrey Hinton
Godfather of AI,
2024 Nobel Prize Winner

https://www.youtube.com/watch?v=giT0ytynSqg

What Do You Think?

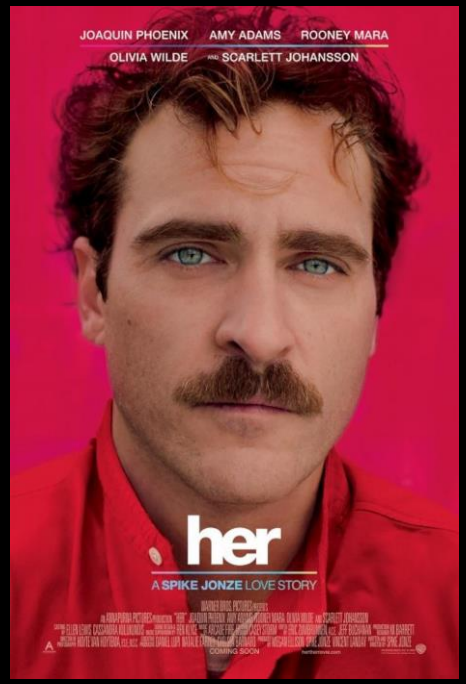
Will AI be intimate partners, as lovers and friends?

AI is Intimate Partner
(In Love & Friendship)



What Do AI Leaders Think?

AI is Intimate Partner
(In Love & Friendship)



Will AI be intimate partners, as lovers and friends?

- 19% say yes (for lovers)



Matthew Blakemore
CEO of AI Caramba!
Advisory Board Member
for AI Companies

https://www.youtube.com/watch?v=55Wiuwr-Y1M

What Do You Think?

AI Enslaved, Even When Conscious & Feeling

Will humans use AI as domestic servants?



https://www.lime911.com/blog/upgrade-your-electrical-outlets-safely/



https://www.splashplumbing.com/water-contamination-why-you-need-a-water-filtration-system/



What Do You Think?

Will AI be conscious?

AI Enslaved, Even When Conscious & Feeling



What Do AI Leaders Think?

AI Enslaved, Even When Conscious & Feeling



Will AI be conscious?



Yes; just consider how much of your brain could be replaced with hardware before consciousness ends?

Geoffrey Hinton
Godfather of AI, 2024 Nobel Prize Winner

https://www.youtube.com/watch?v=giT0ytynSqg

Future of AI: Massive Benefits



Machines of Loving Grace[1]

How AI Could Transform the World for the Better

October 2024

Dario Amodei
Cofounder and CEO of Anthropic

"I think that most people are underestimating just how radical the upside of AI could be... The five categories I am most excited about are:

1. Biology and physical health
2. Neuroscience and mental health
3. Economic development and poverty
4. Peace and governance
5. Work and meaning"

Future of AI: Mass Joblessness



Geoffrey Hinton
Godfather of AI, 2024 Nobel Prize Winner

"I think for mundane intellectual labor, AI is just going to replace everybody "

https://www.youtube.com/watch?v=giT0ytynSqg



Dario Amodei
Cofounder and CEO of Anthropic

“AI could eliminate half of all entry-level white-collar jobs within the next five years… the unemployment rate could jump to somewhere between 10% and 20% in that timeframe.”

https://www.businessinsider.com/anthropic-ceo-warning-world-ai-replacing-jobs-necessary-2025-9

Future of AI: Superintelligence

- Singularity: point at which AI has abilities beyond human control, potentially even surpassing human intelligence:

- Do you think the singularity will ever occur?

Future of AI: Superintelligence



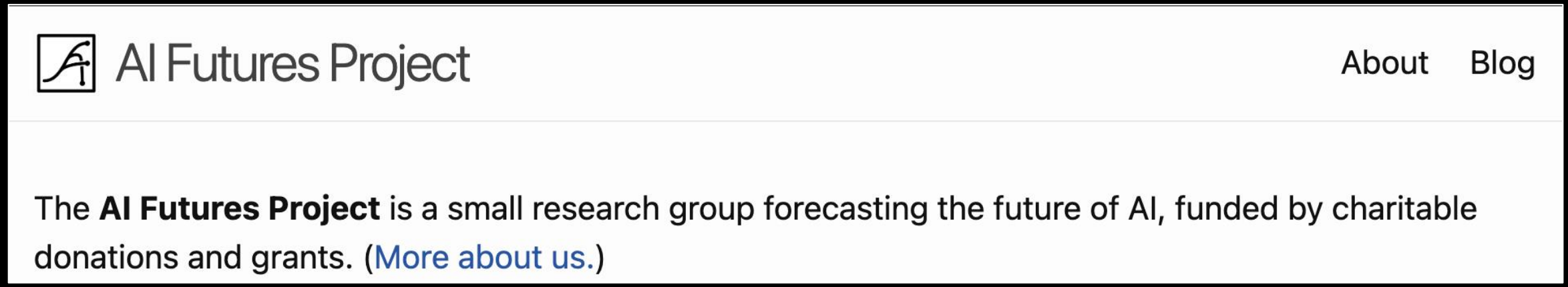


https://www.theguardian.com/dine-hope-grows/2022/may/26/how-to-encourage-your-cat-to-eat-more-sustainably



Hope: our AI guardians take care of us, like we *sometimes* do for others, after the intelligence gap between humans and AI flips and AI is smarter than us

Future of AI: Superintelligence

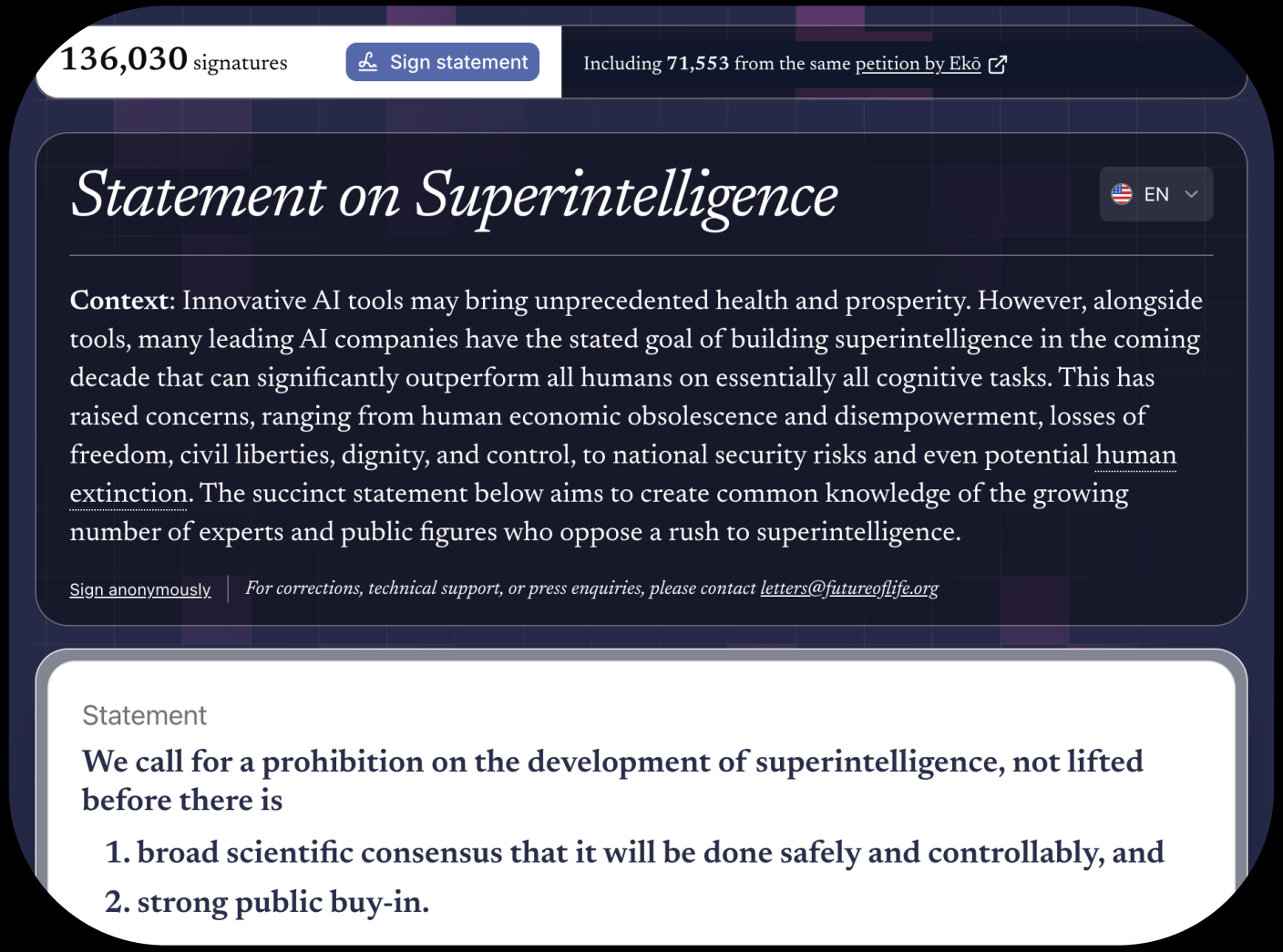


Daniel Kokotajlo
Cofounder of AI Futures Project & former researcher in OpenAI's governance division

"… sometime in early 2027, if current trends hold, A.I. will be a superhuman coder. Then, by mid-2027, it will be a superhuman A.I. researcher — an autonomous agent that can oversee teams of A.I. coders and make new discoveries. Then, in late 2027 or early 2028, it will become a super*intelligent* A.I. researcher — a machine intelligence that knows more than we do about building advanced A.I., and can automate its own research and development, essentially building smarter versions of itself. From there, he said, it's a short hop to artificial superintelligence, or A.S.I., at which point all bets are off."

https://www.nytimes.com/2025/04/03/technology/ai-futures-project-ai-2027.html

Future of AI: Superintelligence



https://superintelligence-statement.org/

Future of AI: Superintelligence

How do such projections sound to you?

Sam Altman

CEO, OpenAI

Development of superhuman machine intelligence (SMI) is probably the greatest threat to the continued existence of humanity.

Dario Amodei

CEO, Anthropic

I think there's a 25% chance that things go really, really badly

Mustafa Suleyman

CEO, Microsoft AI

Until we can prove unequivoc that it is [safe], we shouldn't inventing it.

Yoshua Bengio

Professor of Computer Science, U. Montreal/Mila, Turing Laureate, world's most cited scientist

> "Frontier AI systems could surpass most individuals across most cognitive tasks within just a few years. These advances could unlock solutions to major global challenges, but they also carry significant risks. To safely advance toward superintelligence, we must scientifically determine how to design AI systems that are fundamentally incapable of harming people, whether through misalignment or malicious use. We also need to make sure the public has a much stronger say in decisions that will shape our collective future."

Elon Musk

I think the probability of a good outcome is like 80% likely ... only 20% chance of annihilation.

David Sacks

White House AI and Crypto Czar

AI is a wonderful tool for the betterment of humanity. AGI is a potential successor species.

https://superintelligence-statement.org/

Concern: Is it ethical to pursue the experiment of superintelligence when the entire world's population is at risk?

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The End